Notation : In this sheet, the notation $a \gtrsim b$ means that $a \geq c\,b$ where c is an absolute constant. For asymptotic inequalities that hold up to constant factors, we use the notation $x_N = O(y_N)$. More precisely, $x_N = O(y_N)$ means that there exists a positive constant C such that : $\exists N_0 > 0$, $\forall N \geq N_0$, $x_N \leq C y_N$.

Hoeffding's Inequality

Exercise 1 (Hoeffding's inequality for general bounded r.v. ☕☕)**.** Let $X_1, \dots, X_N$ be independent r.v.. Assume that $X_i \in [m_i, M_i]$ for every i. Then, for any $t > 0$, we have

$$\mathbb{P}\left[\sum_{i=1}^{N}(X_i - \mathbb{E}\,X_i) \,\geq\, t\right] \quad\leq\quad \exp\left(-\frac{2t^2}{\sum_{i=1}^{N}(M_i - m_i)^2}\right).$$

Prove this tail bound, possibly with some absolute constant instead of 2 in the tail.

Exercise 2 (Boosting randomized algorithms ☕☕)**.** Imagine we have an algorithm for solving some decision problem (e.g., is a given number p a prime ?). Suppose that the algorithm makes a decision at random and returns the correct answer with probability $1/2 + \delta$, for some $\delta > 0$, which is just a bit better than a random guess. To improve the performance, we run the algorithm N times and take the majority vote. Show that, for any $\epsilon \in (0, 1)$, the answer is correct with probability $1 - \epsilon$, as long as $N \geq (1/2)\delta^{-2}\ln(\epsilon^{-1})$.

Exercise 3 (Robust estimation of the mean ☕☕☕)**.** Suppose we want to estimate the mean μ of a r.v. X from a sample $X_1, \dots, X_N$ drawn independently from the distribution of X. We want an ϵ-accurate estimate, i.e. one that falls in the interval $(\mu - \epsilon, \mu + \epsilon)$.

1. Show that a sample of size $N \gtrsim \sigma^2/\epsilon^2$ is sufficient to compute an ϵ-accurate estimate with probability at least $3/4$. Here $\sigma^2 = \operatorname{Var} X$.
2. Show that a sample of size $N \gtrsim \log(\delta^{-1})\sigma^2/\epsilon^2$ is sufficient to compute an ϵ-accurate estimate with probability at least $1 - \delta$.

Exercise 4 (Small ball probabilities ☕☕)**.** Let $X_1, \dots, X_N$ be non-negative independent r.v. with continuous distributions. Assume that the densities of X_i are uniformly bounded by 1.

1. Show that the MGF of X_i satisfies $\mathbb{E}\exp(-\lambda X_i) \leq 1/\lambda$, for all $\lambda > 0$.
2. Deduce that, for any $\epsilon > 0$, we have $\mathbb{P}\left[\sum_{i=1}^{N} X_i \leq \epsilon N\right] \leq (e\epsilon)^N$.

Chernoff's Inequality

Exercise 5 (Chernoff's inequality : lower tails ☕☕)**.** Let X_i be independent Bernoulli r.v. with parameters p_i. Consider their sum $S_N = \sum_{i=1}^{N} X_i$ and denote its mean by $\mu = \mathbb{E}\,S_N$. Then, for any $t < \mu$, we have

$$\mathbb{P}\left[S_N \leq t\right] \leq e^{-\mu}\left(\frac{e\mu}{t}\right)^t.$$

Exercise 6 (Poisson tails ☕☕)**.** Let $X \sim \text{Pois}(\lambda)$. Then, for any $t > \lambda$, the tail of X satisfies

$$\mathbb{P}[X \geq t] \leq e^{-\lambda} \left(\frac{e\lambda}{t}\right)^t. \tag{1}$$

1. Prove the Poisson tail bound (1). Comment on it.
2. Show that the bound (1) is quite sharp (i.e., it matches the Poisson distribution up to a negligible factor).

Exercise 7 (Chernoff's inequality : small deviations ☕☕)**.** Show that, in the setting of Exercise 5, for $\delta \in (0, 1]$ we have

$$\mathbb{P}[|S_N - \mu| \geq \delta\mu] \leq 2e^{-c\mu\delta^2},$$

where c is an absolute constant.

Exercise 8 (Poisson distribution near the mean ☕)**.** Let $X \sim \text{Pois}(\lambda)$. Then, for $t \in (0, \lambda]$, the tail of X satisfies

$$\mathbb{P}[|X - \lambda| \geq t] \leq 2\exp\left(-\frac{ct^2}{\lambda}\right).$$

1. Prove the Poisson tail bound above.
2. What do this result and Exercise 6 indicate ?

Exercise 9 (Normal approximation to Poisson ☕☕)**.** Let $X \sim \text{Pois}(\lambda)$. Show that as $\lambda \to \infty$, we have

$$\frac{X - \lambda}{\sqrt{\lambda}} \longrightarrow N(0, 1) \quad \text{in distribution.}$$

Application : Degrees of Random Graphs

Perhaps the most studied model of random graphs is the classical Erdös-Rényi model $G(n, p)$, which is constructed on a set of n vertices by connecting every pair of distinct vertices independently with probability p.

The degree of a vertex in the graph is the number of edges incident to that vertex. The expected degree of every vertex in $G(n, p)$ clearly equals $(n - 1)p := d$. (Check !)

The relatively dense graphs, those where $d \gtrsim \log n$, are almost regular, with high probability, which means that the degrees of all vertices approximately equal d. By contrast, very sparse graphs, those where $d = O(1)$, are far from being regular. The next exercises makes these claims clear.

Exercise 10 (Relatively dense graphs are almost regular ☕☕)**.** There is an absolute constant C such that the following holds. Consider a random graph $G \sim G(n, p)$ with expected degrees $d \geq C \log n$. Then, with high probability (for example, $1 - 1/n$), the following occurs : all vertices of G have degrees between $0.9d$ and $1.1d$.

Exercise 11 (Very sparse graph are far from being regular ☕☕☕)**.** Consider a random graph $G \sim G(n, p)$ with expected degrees $d = O(1)$. We shall assume that the graph size n grows to infinity. Show that with high probability (say, $1 - 1/n$) :

1. all vertices of G have degree $O\left(\frac{\log n}{\log \log n}\right)$.
2. G has a vertex whose degree is at least of order $\frac{\log n}{\log \log n}$.